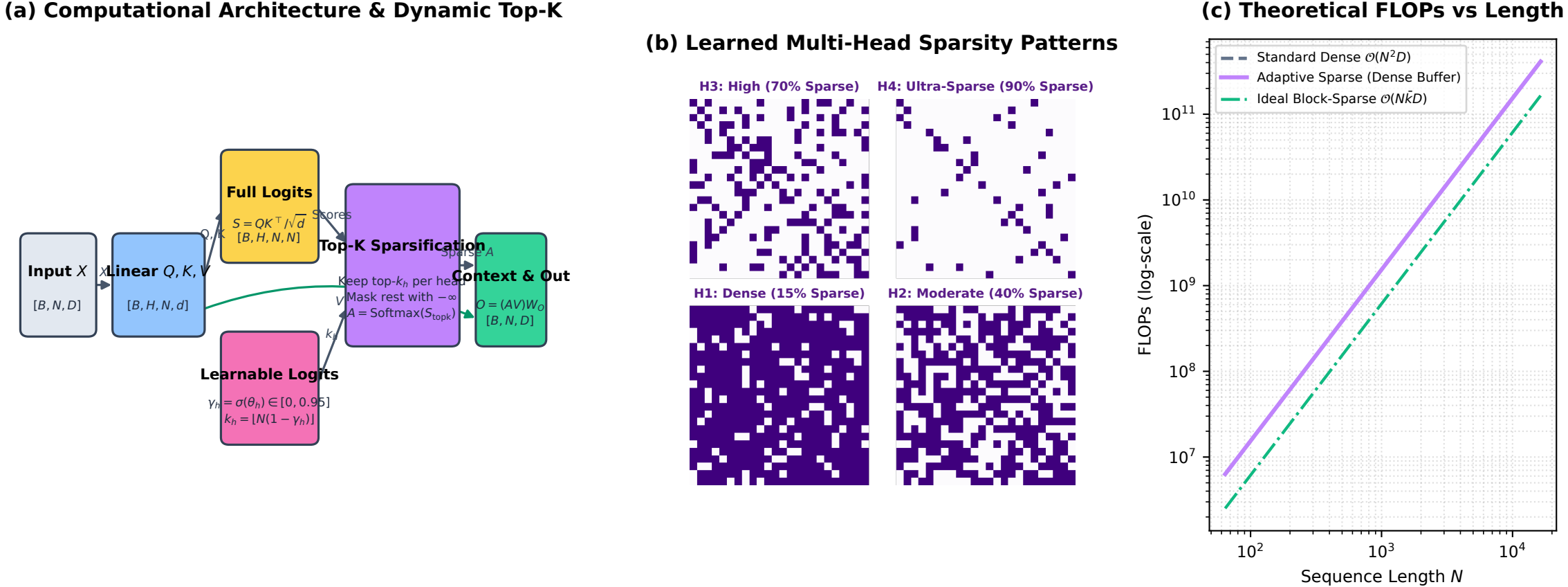
(a) Computational Architecture & Dynamic Top-K
Input X
$[B, N, D]$
Linear Q, K, V
$[B, H, N, d]$
X
Q, K
Full Logits
$S = QK^\top/\sqrt{d}$
$[B, H, N, N]$
Scores
Top-K Sparsification
Keep top-k_h per head
Mask rest with $-\infty$
$A = \mathrm{Softmax}(S_{\mathrm{topk}})$
Sparse A
V
k_h
Learnable Logits
$\gamma_h = \sigma(\theta_h) \in [0, 0.95]$
$k_h = \lfloor N(1-\gamma_h) \rfloor$
Context & Out
$O = (AV)W_O$
$[B, N, D]$
(b) Learned Multi-Head Sparsity Patterns
H3: High (70% Sparse)
H4: Ultra-Sparse (90% Sparse)
H1: Dense (15% Sparse)
H2: Moderate (40% Sparse)
(c) Theoretical FLOPs vs Length
Standard Dense $\mathcal{O}(N^2D)$
Adaptive Sparse (Dense Buffer)
Ideal Block-Sparse $\mathcal{O}(N\bar{k}D)$
FLOPs (log-scale)
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{2}
10^{3}
10^{4}
Sequence Length N